Newton MPM Performance Benchmark Report

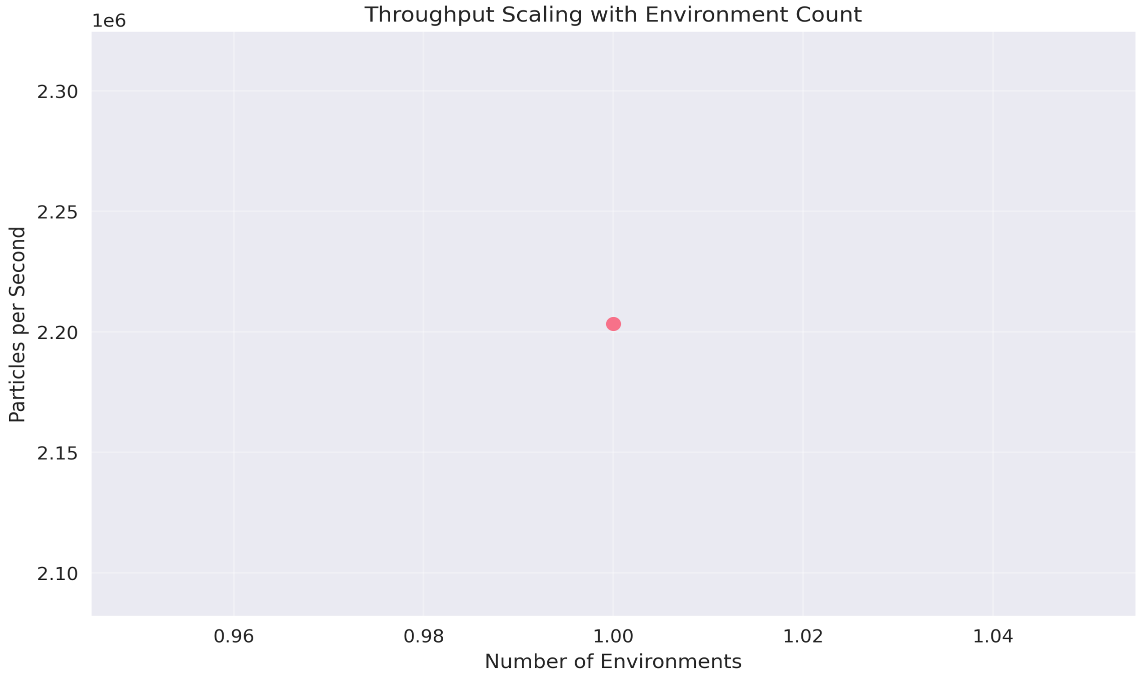
Executive Summary

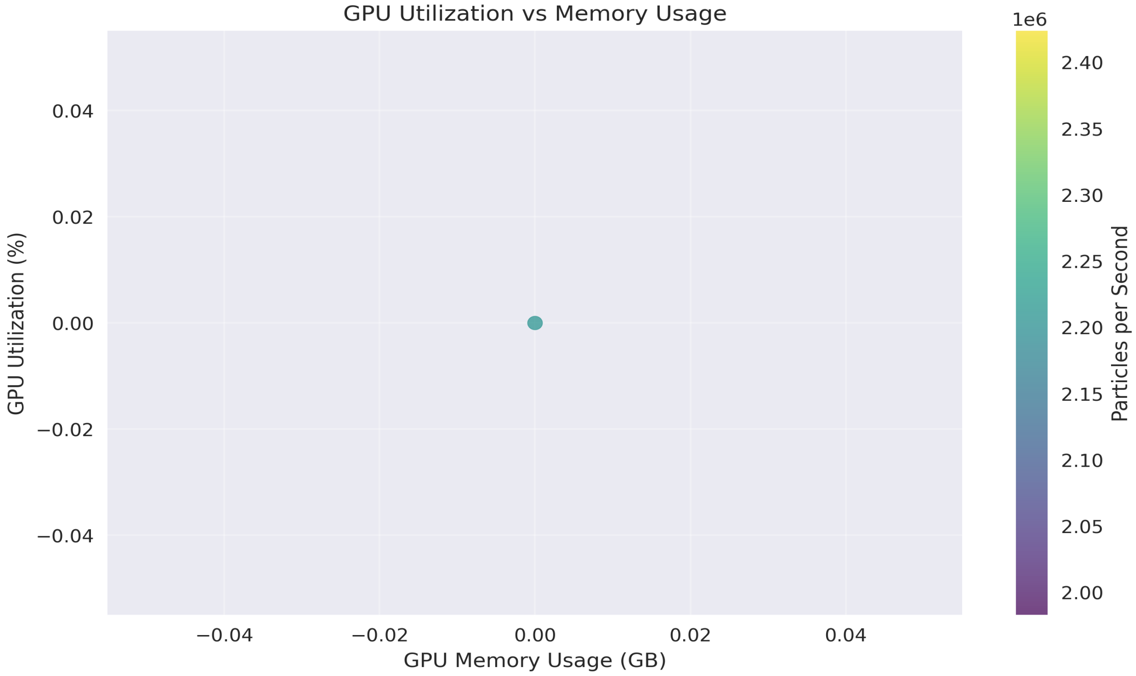
This report presents the performance analysis of Newton's Material Point Method (MPM) particle simulation system. The benchmark was conducted on 2025-08-27 10:30:40 testing 1 different configurations. **Key Findings:**
• Best throughput: 2,203,276 particles/second
• Average GPU utilization: 0.0%
• Memory usage range: 0.0 - 0.0 GB
• Configurations tested: 1

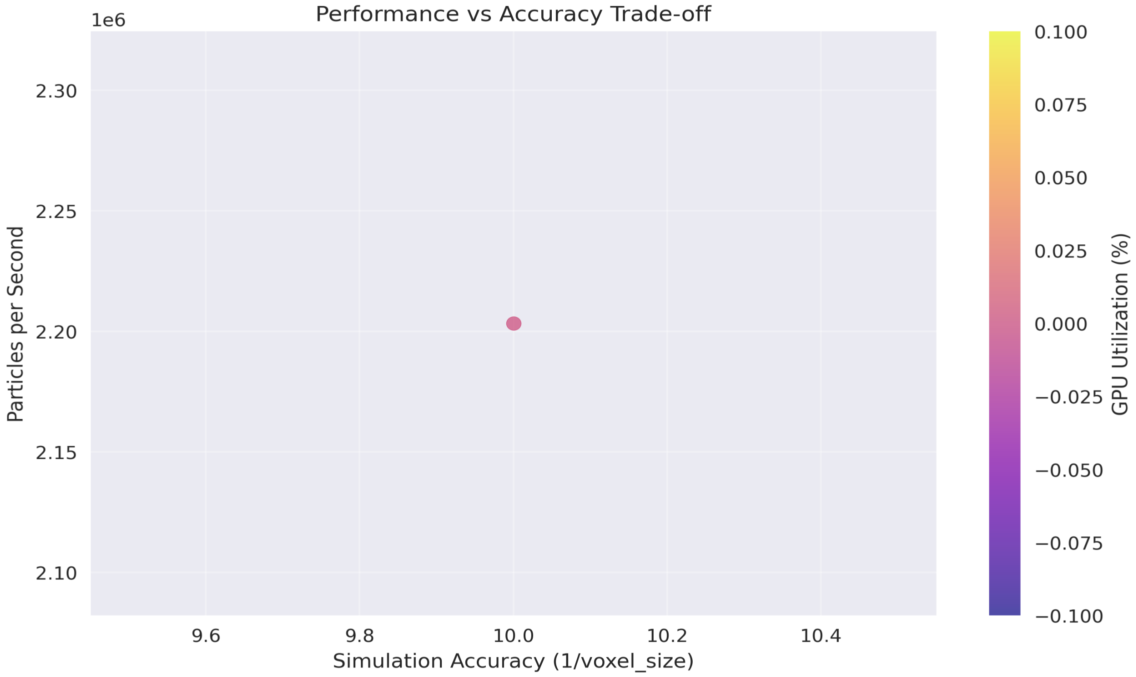
System Configuration

Hardware:
• CPU: AMD Ryzen 9 5900X 12-Core Processor (12 cores)
• Memory: 31.3 GB
• GPU: cuda:0
Software:
• Warp Version: Unknown
• Newton Version: 0.1.1.a2
• GPU Monitoring: Unavailable

Performance Analysis







Optimization Recommendations

1. Best Throughput Configuration: voxel_size=0.1, particles_per_cell=3, achieving 2,203,276 particles/second

2. Best Efficiency Configuration: voxel_size=0.1, particles_per_cell=3, efficiency score: 0

3. Low GPU utilization detected. Consider increasing particle count, reducing voxel size, or adding more environments to better utilize GPU resources.

Detailed Results

Configuration	Particles/sec	GPU Util %	Memory GB
voxel=0.1, envs=1	2,203,276	0.0	0.00